AN R-UNIVERSAL SEMISIMPLE LIE ALGEBRA BECOMES INNER OVER AN IMAGINARY QUADRATIC EXTENSION WITH THE PRESCRIBED STAR-ACTION

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Allowed standard prerequisites (editorial).

- The notion of a Lie algebra being inner over an extension L, and the star-action of Gal(L/Q) compared with the star-action of Gal(C/R). Set up in the author's definitions and notation before the counted statement (source lines 135-216). Fixes the two properties that the statement asserts about the extension L.
- The Tits algebra D(Lie G, Q)(lambda) and the statement that its centre is the minimal extension over which the algebra becomes inner. The author's proposition labelled Beta=D at source lines 393-406, used inside the counted proof. Identifies the centre of the division algebra with the minimal splitting extension and lets the proof conclude that it is imaginary quadratic.

Source reference. Original lemma 2.1: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 416–416. Navigation pointer only: the counted proof invokes the author's equivalence for R-universality in its last paragraph; the equivalence has its own card in this lane and is declared here as a named standard prerequisite.

Date: doi:10.3842/SIGMA.2015.034; CC BY 4.0.

Source reference. Original definition 4: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 432–432. Navigation pointer only: it is the fourth labelled item of that equivalence, cited by the counted proof once.

Source reference. Original definition 3: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 428–428. Navigation pointer only: it is the third labelled item of that equivalence, cited by the counted proof once.

Source reference. Original remark 2.4: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 462–462. Navigation pointer only: the counted proof invokes this remark on the root-lattice case in its closing argument; its content is declared as a named standard prerequisite.

Lemma 0.1. *Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{Q}$. If $\mathfrak{g}$ is $\mathbb{R}$-universal, then $\mathfrak{g}$ is inner over some imaginary quadratic extension L of $\mathbb{Q}$, and the $*$-action of $\mathrm{Gal}(L/\mathbb{Q})$ is the same as the $*$-action of $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$.*

Proposition 0.2 ([3, Corollary 3.5, § 4.2, and Lemma 7.4])**.** *Let L be the center of the algebra $D_{\mathfrak{g},F}(\lambda) := \mathrm{End}_{\mathfrak{g}}({}^{F}\rho_{\lambda})$ (which, by Schur's lemma, is a division algebra). Then:*

1. $\mathrm{Gal}(\overline{F}/L) = \{\sigma \in \mathrm{Gal}(\overline{F}/F) \mid \sigma(\lambda) = \lambda\}$ *(for the $*$-action of $\mathrm{Gal}(\overline{F}/F)$), and*
2. $\beta_{\mathfrak{g},L}(\lambda) = [D_{\mathfrak{g},F}(\lambda)]$*, after identifying $H^2(L;\boldsymbol{\mu})$ with the Brauer group of L.*

Proof. Assume $\mathfrak{g}$ is not inner, for otherwise the desired conclusions are obvious. Let L be the (unique) minimal extension of $\mathbb{Q}$ over which $\mathfrak{g}$ becomes inner, and choose some dominant weight λ that is not fixed by any nontrivial element of $\mathrm{Gal}(L/\mathbb{Q})$. We know that L is the center of $D_{\mathfrak{g},\mathbb{Q}}(\lambda)$ (see Proposition 0.2(1)) and that $D_{\mathfrak{g},\mathbb{Q}}(\lambda)\otimes_{\mathbb{Q}} \mathbb{R}$ is a division algebra (see Lemma 2.1(3)). Since L is not $\mathbb{Q}$, this implies it is an imaginary quadratic extension of $\mathbb{Q}$.

It now suffices to show that every every $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$-invariant dominant weight λ is also $\mathrm{Gal}(L/\mathbb{Q})$-invariant. Suppose not. By replacing λ with an appropriate positive integer multiple, we may assume it is in the root lattice, so $D_{\mathfrak{g},\mathbb{R}}(\lambda) = \mathbb{R}$ (see Remark 2.4). Therefore, we must have $D_{\mathfrak{g},\mathbb{Q}}(\lambda) = \mathbb{Q}$ (see Lemma 2.1(4)). However, this contradicts Proposition 0.2(1), since λ is not $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-invariant. □

References

[1] Morris D., Real representations of semisimple Lie algebras have $\mathbb{Q}$-forms, in Algebraic Groups and Arithmetic, Tata Institute of Fundamental Research, Mumbai, 2004, 469–490.

[2] Prasad G., Rapinchuk A., On the existence of isotropic forms of semi-simple algebraic groups over number fields with prescribed local behavior, *Adv. Math.* **207** (2006), 646–660.

[3] Tits J., Représentations linéaires irréductibles d'un groupe réductif sur un corps quelconque, *J. Reine Angew. Math.* **247** (1971), 196–220.